IoT Homework – Exercise 1

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1 Probability Mass Function

The number of vibration peaks detected in each 10-second window follows a Poisson distribution with parameter:

$$\lambda = 0.4$$

The probability mass function of a Poisson random variable is:

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Therefore:

$$P(r = r_0) = P(k = 0) = e^{-0.4} = 0.6703$$

$$P(r = r_1) = P(k = 1) = 0.4e^{-0.4} = 0.2681$$

$$P(r = r_2) = P(k > 1)$$

$$P(r = r_2) = 1 - P(k = 0) - P(k = 1)$$

$$P(r = r_2) = 1 - 0.6703 - 0.2681 = 0.0616$$

Thus, the PMF is:

$$P(r = r_0) = 0.6703$$

$$P(r = r_1) = 0.2681$$

$$P(r = r_2) = 0.0616$$

2 Output Rates

The reporting period is:

$$T_{report} = 10s$$

If no vibration peaks are detected, the payload size is:

$$512B$$

Therefore:

$$r_0 = \frac{512 \cdot 8}{10}$$

$$r_0 = 409.6\ bit/s$$

If one vibration peak is detected, the payload size is:

$$2KB = 2000B$$

Thus:

$$r_1 = \frac{2000 \cdot 8}{10}$$

$$r_1 = 1600 \ bit/s$$

If more than one vibration peak is detected, the payload size is:

$$4KB = 4000B$$

Thus:

$$r_2 = \frac{4000 \cdot 8}{10}$$

$$r_2 = 3200 \ bit/s$$

3 Beacon Interval

According to the standard IEEE 802.15.4 formulation, the Beacon Interval is computed from the minimum required rate.

The minimum output rate is:

$$r_{min} = 409.6 \ bit/s$$

Given:

$$L = 128B$$

the Beacon Interval is:

$$BI = \frac{L \cdot 8}{r_{min}}$$

Therefore:

$$BI = \frac{128 \cdot 8}{409.6}$$

$$BI = 2.5s$$

This means that the reporting period of 10 seconds is divided into:

$$\frac{10}{2.5} = 4$$

Beacon Intervals.

4 CFP Slot Assignment

The system operates with IEEE 802.15.4 in beacon-enabled mode using CFP only.

Given:

$$R = 250kbps$$

$$L = 128B$$

The slot duration is:

$$T_s = \frac{L \cdot 8}{R}$$

$$T_s = \frac{128 \cdot 8}{250000}$$

$$T_s = 0.004096s$$

$$T_s = 4.096ms$$

The CFP is dimensioned according to the worst-case traffic scenario.

The largest payload generated by one node is:

$$4000B$$

The number of packets required over the full reporting period is:

$$N_{packets} = \frac{4000}{128}$$

$$N_{packets} = 31.25$$

Approximating to the next integer:

$$N_{packets} = 32$$

Since the reporting period spans 4 Beacon Intervals, the required number of slots per BI is:

$$N_{slots,node} = \frac{32}{4}$$

$$N_{slots,node} = 8$$

The monitoring system contains 3 vibration monitoring nodes.

Therefore:

$$N_{CFP} = 3 \cdot 8$$

$$N_{CFP} = 24$$

Thus, the CFP slot assignment is:

$$8 \; slots/node$$

and the total number of CFP slots is:

$$N_{CFP} = 24$$

5 Active and Inactive Time

The active time includes:

- one beacon slot;
- the CFP slots.

Therefore:

$$N_{active} = 24 + 1 = 25$$

The active time is:

$$T_{active} = 25 \cdot 4.096ms$$

$$T_{active} = 102.4ms$$

The inactive time is:

$$T_{inactive} = BI - T_{active}$$

$$T_{inactive} = 2500ms - 102.4ms$$

$$T_{inactive} = 2397.6ms$$

$$T_{inactive} = 2.3976s$$

6 Duty Cycle

The duty cycle is computed as:

$$DC = \frac{T_{active}}{BI}$$

$$DC = \frac{102.4}{2500}$$

$$DC = 0.04096$$

$$DC = 4.096\%$$

7 Additional Nodes

The maximum allowed duty cycle is:

$$DC_{max} = 10\%$$

The active time is:

$$T_{active} = (1 + 8M)T_s$$

where M is the number of vibration monitoring nodes.
Imposing the duty-cycle constraint:

$$\frac{(1 + 8M) \cdot 4.096}{2500} < 0.1$$

$$1 + 8M < 61.035$$

$$8M < 60.035$$

$$M < 7.50$$

Therefore, the maximum integer number of nodes is:

$$M_{max} = 7$$

The system already contains 3 nodes.
Therefore:

$$\Delta M = 7 - 3$$

$$\Delta M = 4$$

Hence, 4 additional vibration monitoring nodes can be added while keeping the duty cycle below 10%.

$$\Delta M = 4 \; additional \; nodes$$